Algorithm 1 Meta-evolution

Inputs: baseline; G generations; O offspring per generation; P population size; tasks T; N_{init} initial seeds; B TTTS seeds per generation; N_{reeval} uniform reevaluation seeds; N_{val} validation seeds; reevaluation `strategy` $\in \{\texttt{TTTS}, \texttt{uniform}\}$.

`population` $\leftarrow$ [`baseline`]
`archive` $\leftarrow \{\}$
for $g = 1, \ldots, G$ **do**
  for $o = 1, \ldots, O$ **do** ▷ *Sample offspring*
    `parents` $\sim$ SampleParents(`population`)
    `prompt` $\sim$ Prompt($T, g,$ `parents`)
    $x_o \sim$ LLM(`prompt`)
    $\hat{\mu}_{x_o} \leftarrow$ Evaluate(x_o, T, N_{init})
    $n_{x_o} \leftarrow N_{\text{init}}$
    `archive` $\leftarrow$ `archive` $\cup \{x_o\}$
  if `strategy` $=$ `TTTS` **then** ▷ *Reevaluate*
    $\psi_x \leftarrow$ top-two Thompson probability of x under $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$
    Draw B members from `archive` with probability $\propto \psi_x$
  else if `strategy` $=$ `uniform` **then**
    Draw each new member of `population` N_{reeval} times
  Evaluate each drawn member on new seeds and update $\hat{\mu}_x$, n_x
  `population` $\leftarrow \{\arg\max_{x \in \texttt{archive}} \hat{\mu}_x(t) \mid t \in T\}$ ▷ *MAP-Elites, one niche per task*
  $x_g^* \leftarrow \arg\max_{x \in \texttt{population}} \hat{\mu}_x$ ▷ *Validate generation's best*
  `val`$[x_g^*] \leftarrow$ Evaluate($x_g^*, T_{\text{val}}, N_{\text{val}}$)
return $\arg\max_x$ `val`$[x]$